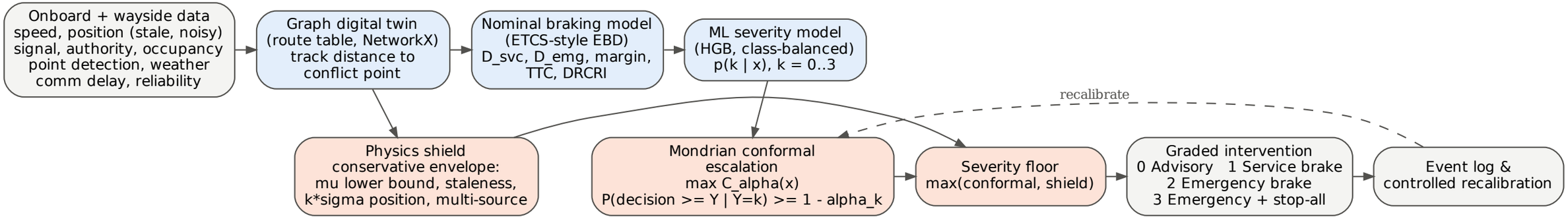
Onboard + wayside data
speed, position (stale, noisy)
signal, authority, occupancy
point detection, weather
comm delay, reliability
Graph digital twin
(route table, NetworkX)
track distance to
conflict point
Nominal braking model
(ETCS-style EBD)
D_svc, D_emg, margin,
TTC, DRCRI
ML severity model
(HGB, class-balanced)
p(k | x), k = 0..3
recalibrate
Physics shield
conservative envelope:
mu lower bound, staleness,
k*sigma position, multi-source
Mondrian conformal
escalation
max C_alpha(x)
P(decision >= Y | Y=k) >= 1 - alpha_k
Severity floor
max(conformal, shield)
Graded intervention
0 Advisory 1 Service brake
2 Emergency brake
3 Emergency + stop-all
Event log &
controlled recalibration